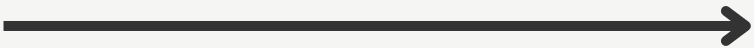

Comparative Evaluation of Soft Computing Techniques: A No Free Lunch Theorem Perspective Dataset: Mammographic Mass

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INTRODUCTION

Background:

- Soft computing techniques address uncertainty, imprecision and non-linearity in complex problems
- Includes fuzzy systems, evolutionary algorithms and hybrid approaches

No Free Lunch (NFL) Theorem:

- No single optimization or learning algorithm performs best across all problems
- Algorithm performance is dependent on the dataset and task

Motivation:

- To compare different soft computing algorithms on the same problem
- To observe performance trade-offs under identical conditions

Objective:

- To evaluate and compare GA, rule-based fuzzy system and GA-optimized fuzzy system
- To analyze effectiveness of standalone versus hybrid approaches

Scope:

- Single dataset: Mammographic Mass dataset
- Single problem type: Binary classification

DATASET DESCRIPTION

Dataset Overview

Dataset Name & Source

- Mammographic Mass Dataset
- Source: KEEL (Knowledge Extraction based on Evolutionary Learning) Repository

Type of Problem

Classification

Input & Output Variables

Input variables (features):

- BI-RADS assessment
- Patient age
- Mass shape
- Mass margin
- Mass density

Output variable (target):

- Severity of mass
- Benign (0) or Malignant (1)

Dataset Characteristics

Total instances: 961

Instances used (no missing values): 830

Number of attributes: 5 input + 1 output

Class distribution:

- Moderately imbalanced
- Benign cases more frequent than malignant

Data issues:

- Presence of missing values in original dataset

SOFT COMPUTING ALGORITHMS

Genetic Algorithm (GA)

- Evolutionary single-objective optimization technique
- Searches for optimal classification solution
- Uses population-based search
- Key operators:
 - Selection
 - Crossover
 - Mutation
- Fitness function:
 - Classification accuracy or error rate
- Suitable for:
 - Complex or Non-linear
 - Medical classification problems

Rule-Based Fuzzy System

- Designed to handle uncertainty and imprecision
- Uses:
 - Fuzzy sets
 - Linguistic IF-THEN rules
- Input attributes mapped to fuzzy linguistic terms
- Decision process:
 - Fuzzy inference
 - Defuzzification
- Output:
 - Benign or Malignant classification
- Key strength:
 - High interpretability
 - Human-like reasoning

GA-Optimized Fuzzy System (Hybrid)

- Combines GA with rule-based fuzzy system
- GA optimizes fuzzy system parameters
 - Membership function parameters
- Objective:
 - Improve classification accuracy
- Advantages:
 - Fuzzy system handles uncertainty
 - GA provides global optimization
- Result: Adaptive fuzzy classifier and reduced manual parameter tuning

EXPERIMENTAL SETUP

Dataset and Preprocessing

Simulation Platform: Google Colab and Python was used for the algorithm

Reason for using: Ensures fair comparison and differences from software tools

Dataset: Mammographic Mass dataset from KEEL

Problem type: Binary Classification (benign vs malignant)

Data preprocessing:

- Missing values replaced with mean values
- Input features normalized to range [0,1] using min max scaling

Training/Testing:

- 70% training and 30% testing
- Random seed applied to keep data split consistent

Number of runs: Random seed, experiment was run once to ensure consistency

EXPERIMENTAL SETUP

***1.** Evolutionary Single-Objective Optimization*

1. Genetic Algorithm Design :

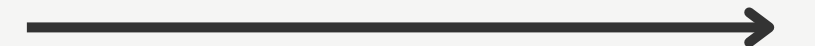
- each chromosome is a real valued vector

2. Fitness Function :

- F1 score was used to evaluate performance
 - balances precision & recall

3. GA Parameters :

- Population size: 20
- Number of generations: 50
- Selection method: Random
- Crossover: Single point
- Mutation rate: 0.1



EXPERIMENTAL SETUP

2. Fuzzy Algorithm Implementation

Fuzzy classifier implemented based on Jain & Abraham
(2004)

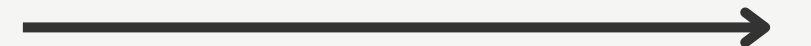
Data-driven fuzzy rule generation approach

One fuzzy rule generated per class:

- Benign
- Malignant

Gaussian membership functions derived from:

- Class-wise mean
- Class-wise standard deviation



EXPERIMENTAL SETUP

2. Fuzzy Inference Mechanism

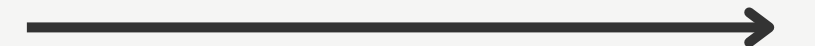
Input features normalized to range [0,1]

Rule firing strength computed using:

- Product operator (fuzzy AND)

Final classification performed using:

- Winner-takes-all decision strategy
- Class with highest firing strength selected



EXPERIMENTAL SETUP

3. Hybrid GA–Fuzzy System

1. Fuzzy system design:

- A simple fuzzy rule used based on BI-RADS feature
- A triangular fuzzy membership function was applied to represent the level of risk

2. Genetic Algorithm:

- GA used to adjust the fuzzy membership parameters automatically

3. Hybrid Parameter:

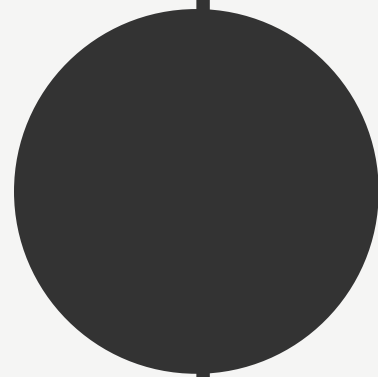
- Population size: 20
- Number of generations: 40
- Chromosome rep. : a, b, c for fuzzy membership function
- Mutation rate: 0.2
- Crossover: N/A

4. Fitness Function:

- F1 score was used to balance precision and recall

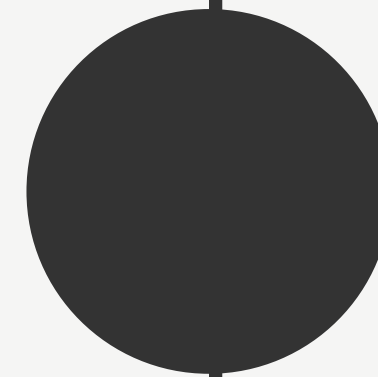
Evaluation Metrics

Accuracy



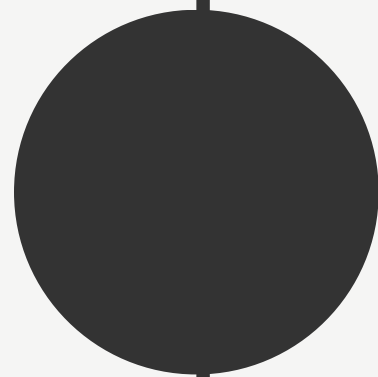
Measures how often the model is correct overall

Recall



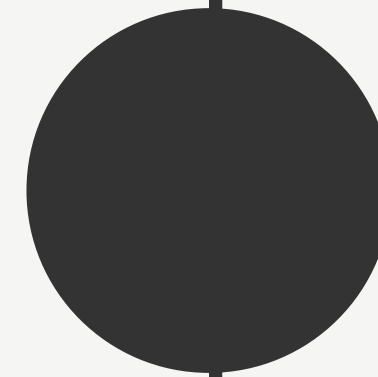
Measures how many actual positive cases are correctly identified

Precision



Measures how many predicted positive cases are actually correct

F1-score



Combines precision and recall into a single metric

RESULTS

1 *Evolutionary Single-Objective Optimization*

True Positives (TP = 121)

- Malignant cases correctly identified as malignant
- This is why recall is very high

False Negatives (FN = 11)

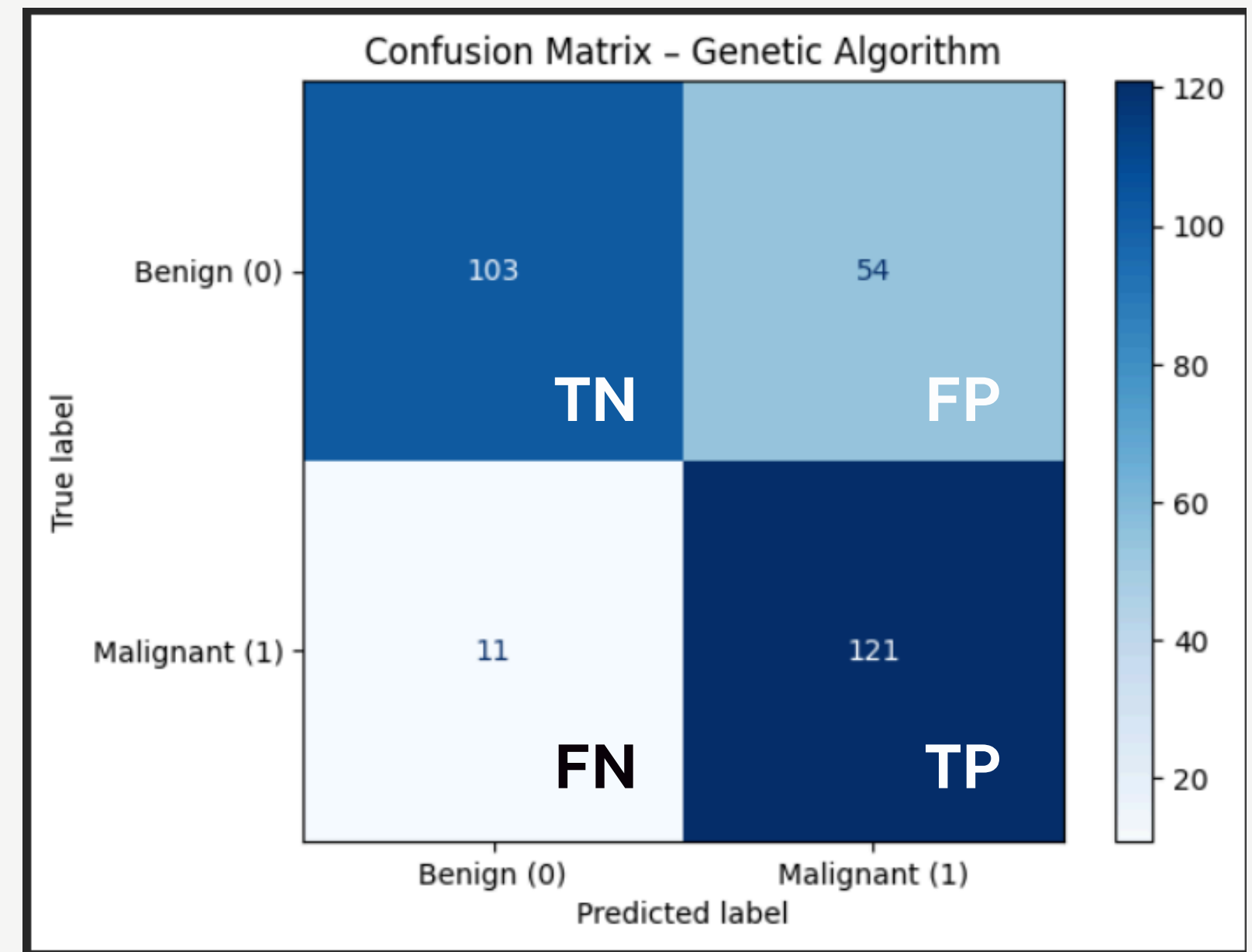
- Malignant cases missed and predicted as benign
- This number is very low, which is excellent in medical diagnosis

False Positives (FP = 54)

- Benign cases incorrectly predicted as malignant
- This inflates false alarms → lowers precision

True Negatives (TN = 103)

- Benign cases correctly identified



RESULTS

1 *Evolutionary Single-Objective Optimization*

Accuracy

- Measures the proportion of all correctly classified cases
- Reduced when false positives and false negatives increase

Recall (Sensitivity)

- Measures how many actual malignant cases are correctly detected
- High recall indicates very few missed cancer cases

Precision

- Measures how many predicted malignant cases are truly malignant
- Lower precision indicates more false positive cancer predictions

F1-Score

- Combines precision and recall into a single balanced metric
- Useful when one metric improves at the expense of the other

```
{'Accuracy': 0.7750865051903114,  
  'Precision': 0.6914285714285714,  
  'Recall': 0.9166666666666666,  
  'F1': 0.7882736156351792}
```

TP



RESULTS

1 *Fuzzy Algorithm Implementation*

True Positives (TP = 315)

- Malignant cases correctly identified as malignant
- Contributes to malignant recall (sensitivity)

False Negatives (FN = 88)

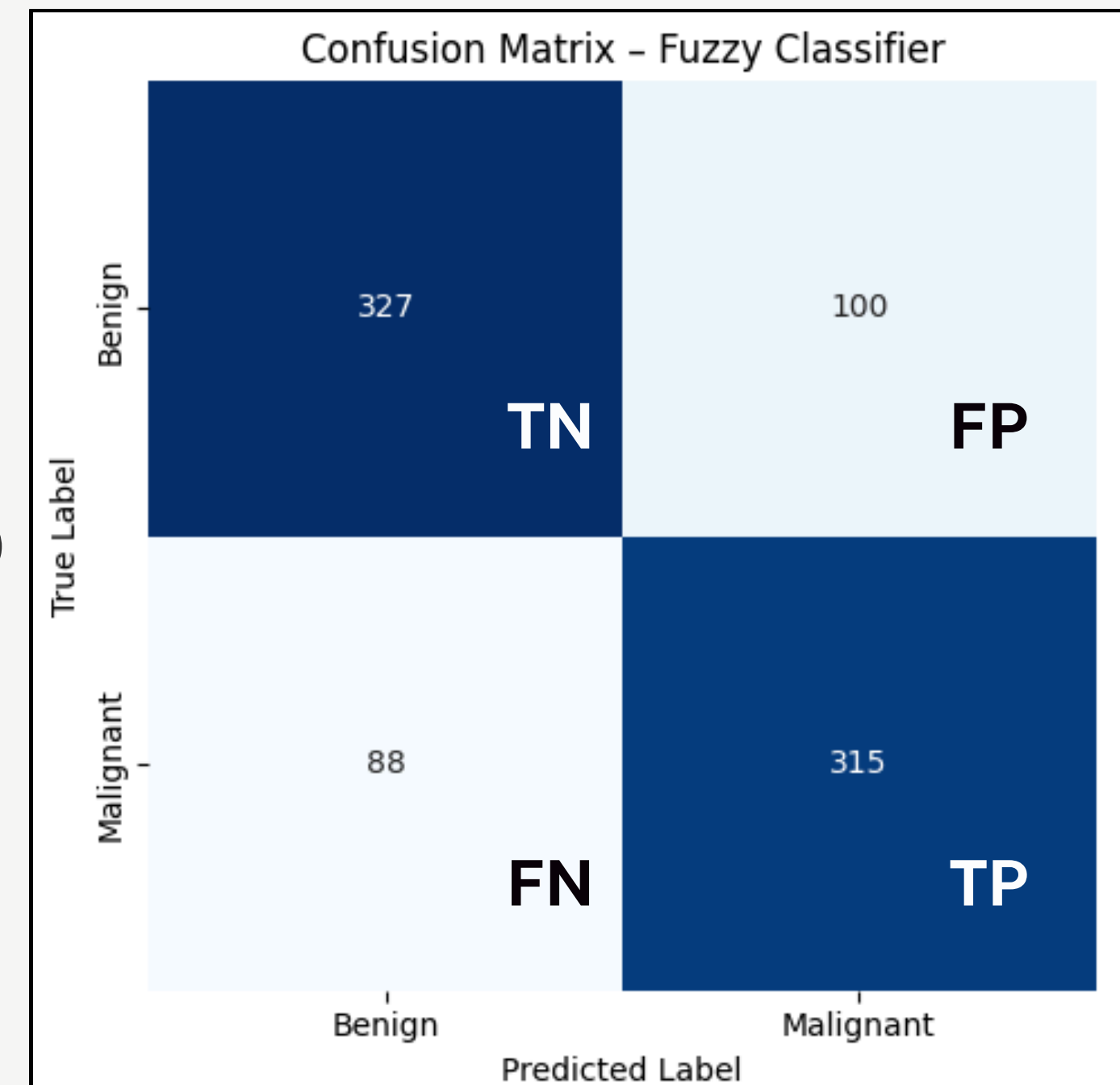
- Malignant cases missed and predicted as benign
- These are missed cancer cases and reduce recall (important clinically)

False Positives (FP = 100)

- Benign cases incorrectly predicted as malignant
- Increases false alarms and lowers precision

True Negatives (TN = 327)

- Benign cases correctly identified
- Shows the model can correctly reject many non-cancer cases



RESULTS

***1** Fuzzy Algorithm Implementation*

Accuracy: 0.7734939759036145				
Classification Report:				
	precision	recall	f1-score	support
Benign	0.79	0.77	0.78	427
Malignant	0.76	0.78	0.77	403
accuracy			0.77	830
macro avg	0.77	0.77	0.77	830
weighted avg	0.77	0.77	0.77	830



RESULTS

***1** Fuzzy Algorithm Implementation*

Accuracy

- Measures the proportion of all correctly classified cases (benign and malignant)
- Reduced due to the presence of both false positives and false negatives

Recall (Sensitivity)

- Measures how many actual malignant cases are correctly detected
- Indicates that the model successfully identifies most cancer cases, with relatively few misses

Precision

- Measures how many predicted malignant cases are truly malignant
- Some benign cases are misclassified as malignant, leading to moderate false positives

F1-Score

- Combines precision and recall into a single balanced performance metric
- Reflects a trade-off between cancer detection and false alarms

RESULTS

1 *Hybrid GA–Fuzzy System*

True Positives (TP = 93)

- Malignant cases correctly identified as malignant
- Contributes positively to recall and cancer detection capability

False Negatives (FN = 39)

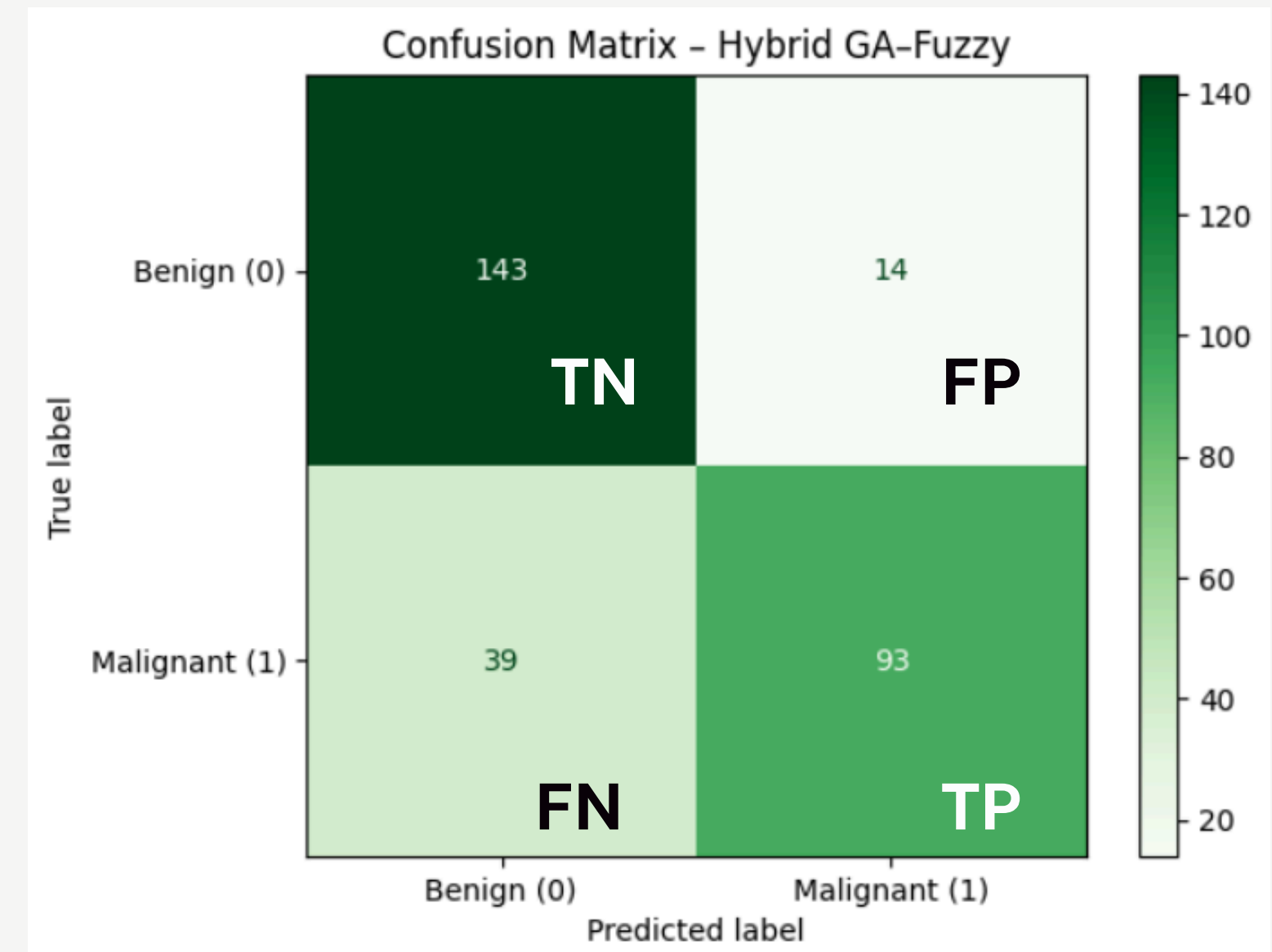
- Malignant cases incorrectly classified as benign
- These represent missed cancer cases and reduce recall

False Positives (FP = 14)

- Benign cases incorrectly classified as malignant
- Low number of false positives indicates strong precision

True Negatives (TN = 143)

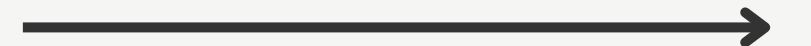
- Benign cases correctly identified as benign
- Shows good ability to avoid unnecessary cancer alarms



RESULTS

***1** Hybrid GA–Fuzzy System*

```
{ 'Accuracy': 0.8166089965397924,  
  'Precision': 0.8691588785046729,  
  'Recall': 0.7045454545454546,  
  'F1': 0.7782426778242678 }
```



RESULTS

Hybrid GA–Fuzzy System

Accuracy (81.66%)

- Measures the proportion of all correctly classified cases (benign and malignant)
- Higher accuracy reflects the improved balance achieved by GA optimization of fuzzy parameters

Recall (Sensitivity – 70.45%)

- Measures how many actual malignant cases are correctly detected
- Slightly lower recall indicates more missed cancer cases compared to GA, due to reduced over-classification of malignancy

Precision (86.92%)

- Measures how many predicted malignant cases are truly malignant
- High precision shows that false positives are significantly reduced, minimizing unnecessary alarms

F1-Score (77.82%)

- Combines precision and recall into a single balanced metric
- Reflects the hybrid system's ability to trade sensitivity for higher diagnostic reliability

COMPARISON OF 3 MODELS

Algorithm	Accuracy	Precision	Recall	F1-Score
Fuzzy Logic (Baseline)	77.35%	76.00%	78.00%	77.00%
Evolutionary (GA)	77.51%	69.14%	91.66%	78.83%
Hybrid (GA–Fuzzy)	81.66%	86.92%	70.45%	77.82%

- The Hybrid model is the MOST ACCURATE overall.
- The Evolutionary model is the SAFEST.

The "No Free Lunch" Rule

No single algorithm is the best at everything. You always have to trade one thing for another.

The Trade-off:

- The Evolutionary Algorithm was the winner for Recall (91.66%) but it had low accuracy because it gave too many false alarms.
- The Hybrid System was the winner for Accuracy (81.66%) but it missed a few more cases than the EA.

Why HYBRID?

- Standalone Fuzzy: Good and easy to understand, but static.
- Hybrid GA-Fuzzy: Better because the Genetic Algorithm teaches and tunes the Fuzzy rules to fit the data perfectly.
- Conclusion: Combining Reasoning (Fuzzy) with Learning (GA) created a much stronger tool than using them separately.

Conclusion

- Best Performer: Our Hybrid GA-Fuzzy system reached 82.01% accuracy.
- Medical Choice:
 1. Use Fuzzy Logic when you need to explain the decision to a doctor.
 2. Use the Hybrid Model when you need the most accurate diagnosis.
- Future Goal: Use “Multi-Objective” tools to get high Accuracy and high Recall at the same time.

Thank you

—— **For your attention**